

Programme	B. Sc. Computer Science				
Course Title	Object Oriented Programming (Java)				
Type of Course	Major				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Knowledge in basic programming 2. Knowledge in OOP Concepts				
Course Summary	The aim of this course is to provide students with an understanding of the basic concepts in Java programming. This course will help students create GUI applications in Java and establish database connectivity.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To understand the concepts and features of Object Oriented Programming(OOPs)	U	C	
CO2	To practice programming in Java	Ap	P	
CO3	To learn java's exception handling mechanism, I/O operations and multithreading.	Ap	P	
CO4	To learn java's O operations and multithreading.	Ap	P	
CO5	Implement programs using Java Database Connectivity	Ap	P	
CO6	Students will be capable of developing Graphical User Interface (GUI) applications using Swing, understanding layout management, and implementing basic event handling.	Ap	P	
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks (70)
I	Review of OOPs and Introduction to Java		17	20
	1	Overview of OOPs Concept	1	
	2	History of Java and Java Virtual Machine	1	
	3	Basic Structure of Java Programming : Data Types, Operators, Expression and Control Statement	2	
	5	Arrays and String: One Dimensional Array, Multidimensional Array, String Operations	2	
	6	Scanner, Type Conversion and Casting	2	
	7	Introduction to Class and Objects: Definition of Class and Objects, Access Modifier	2	
	8	Constructor and Inheritance: Types of Constructors, Types of Inheritance, use of extends, super, final, this keyword	3	
	9	Method Overriding, Method Overloading and Dynamic Method Dispatch : Programming implementation of Method Overriding and Overloading	2	
	10	Interface, Abstract Class and Packages ; Programming implementation of Interface, Abstract class and Packages	2	
II	Exception and I/O Operations		8	15
	11	Exception: Baic Concept of exception and Exception Hierarchy	2	
	12	Managing Exception: Use of try....catch finally blocks, throw and throws keyword	2	
	13	Managing Input/Output files in Java : Importance of I/O Operations, BufferedInputStream, BufferedOutputStream	2	
	14	File Operations : Programming implementation of FileInputStream, FileOutputStream, FileReader, FileWriter	2	
III	Multithreading and Database Connectivity		9	20
	15	Thread : Concept of Thread and Thread state	2	
	16	Programming Implementation of Thread : Using extending thread class and Runnable interface, Thread Priorities	2	

	17	Database Programming : Basic Concept of Database and JDBC Driver, Connecting with Database	2	
	18	Querying Database: Programming implementation of creating table, insert and update values to the table using	3	
		preparedStatement, Statement object and querying the values using ResultSet and ResultSetMetadata		
IV	GUI Programming		11	15
	19	Introduction to GUI Application : AWT Basics, Introduction to IDE	2	
	20	Swing Programming : Introduction of Model-View-Controller Pattern	2	
	21	Introduction to layout Management : Fundamental controls used in SWING	4	
	22	Event Handling : Basic Knowledge of Event Handling(Event Class and Event Listener)	3	
V	Hands-on Programming in Java(Using IDE NetBeans, Eclipse, VSCode):		30	30
	Practical Applications, Case Study and Course Project			
	1	Implement the following:		
		1. String and Arrays:	20	
		Write a program to perform various String operations in Java(Hint: charAt, substring, concat, equals,, isEmpty..)		
		Write a program to implement Multi-Dimensional Array(Hint : Matrix multiplication)		
		2. Object Oriented Programming Concept:		
		Write a program to implement the concept of class and object.(Hint: Complex Number addition)		
		Write a program to demonstrate the order in which constructors are invoked in multilevel inheritance.		
		Write a program to implement method overloading		
		Write a program to implement method overriding.		
		3. Exception Handling and Multithreading:		
		Write a program to implement try...catch, finally block (Hint: Arithmetic and ArrayOutOfBoundException)		

		Write a multi thread java program for displaying odd numbers and even numbers up to a limit (Hint :Create thread by inheriting Thread class).		
		Write a multi thread java program for displaying odd numbers and even numbers up to a limit (Hint : Implement thread using Runnable interface).		
		4. GUI Application with Database:		
		Write a swing program to track mouse & key events		
		Write a swing program to fetch data from TextFiled and display it in Label		
		Write a swing program to perform form validation		
		Write a swing program to display data in tabular form		
		Write a simple login program without database connectivity		
		Write a swing program to create a registration form (Hint : Create table student in any database and link the registration form with database using JDBC)		
	2	Case Study	2	
	3	Project: Build a application for shop management system (Eg: Admin Login, Product registration, stock management, product selling, employee salary)	8	

Text Book :

1. Herbert Scheldt, Java: The Complete Reference, 12th Edition, Tata McGraw-Hill Edition, ISBN: 9781260463415.

References :

1. C. Thomas Wu, An introduction to Object-oriented programming with Java, 5e, McGraw-Hill, 2009.
2. Y. Daniel Liang, Introduction to Java programming, Comprehensive Version, 10e, Prentice Hall India, 2013.
3. K. Arnold, J. Gosling, David Holmes, The JAVA programming language, 4e, Addison- Wesley, 2005.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
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CO 1	1	-	3	3	-	-						
CO 2	1	-	3	3	-	-						
CO 3	-	-	3	3	2	3						
CO 4	-	-	2	3	-	-						
CO 5	-	-	3	3	2	3						
CO 6			3	3	3							

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓